

Chit 10 — Database Trigger: Row Level BEFORE Trigger

Table Creation

```
CREATE TABLE Library (  
  Book_Id    NUMBER PRIMARY KEY,  
  Book_Name  VARCHAR2(100),  
  Author     VARCHAR2(50),  
  Qty        NUMBER  
);  
  
CREATE TABLE Library_Audit (  
  Book_Id    NUMBER,  
  Book_Name  VARCHAR2(100),  
  Author     VARCHAR2(50),  
  Qty        NUMBER,  
  Action     VARCHAR2(10),  
  Changed_On DATE DEFAULT SYSDATE  
);  
  
INSERT INTO Library VALUES (1,'DBMS','Korth',5);  
INSERT INTO Library VALUES (2,'OS','Galvin',3);  
COMMIT;
```

BEFORE UPDATE Trigger

```
CREATE OR REPLACE TRIGGER trg_library_before_update  
BEFORE UPDATE ON Library  
FOR EACH ROW  
BEGIN  
  INSERT INTO Library_Audit  
    (Book_Id, Book_Name, Author, Qty, Action, Changed_On)  
  VALUES  
    (:OLD.Book_Id, :OLD.Book_Name, :OLD.Author,  
     :OLD.Qty, 'UPDATE', SYSDATE);  
END;  
/
```

BEFORE DELETE Trigger

```
CREATE OR REPLACE TRIGGER trg_library_before_delete  
BEFORE DELETE ON Library  
FOR EACH ROW  
BEGIN  
  INSERT INTO Library_Audit  
    (Book_Id, Book_Name, Author, Qty, Action, Changed_On)  
  VALUES  
    (:OLD.Book_Id, :OLD.Book_Name, :OLD.Author,  
     :OLD.Qty, 'DELETE', SYSDATE);  
END;  
/
```

Test Triggers

```
UPDATE Library SET Qty = 10 WHERE Book_Id = 1;  
DELETE FROM Library WHERE Book_Id = 2;  
SELECT * FROM Library_Audit;
```

